



School of Computing and Engineering Sciences
Bachelor of Business Information Technology
2022/2023 Academic Year
BBT1201 - Database Systems (FT)
End of Semester Examination

Sep. 2022*1

Time: 2 Hours

Instructions: Answer question ONE and any other TWO questions of your choice.

Auxiliary Examination Information

Consider the relation Members, and use it to answer the questions that follow.

Relation Member

Field Name	Data Type	Constraints
Mem_No	Number(10)	Primary Key
Mem_Fname	Varchar(15)	Not Null
Mem_Lname	Varchar(15)	Not Null
Mem_Sex	Varchar(1)	Not Null, Assumes the Values "F" for Female OR "M" for Male.
Mem_Email	Varchar(20)	Not Null, Check Email Format, Unique
Mem_DOB	Date	Not Null
Mem_Acc_Balance	Number(10,2)	Not Null, Check >=0

Member Data

Mem_No	Mem_Fname	Mem_LName	Mem_Sex	Mem_Email	Mem_DOB	Mem_Acc_Bal
907	Paul	Otiato	M	lopo78@gmail.com	1-Jan-2000	0
998	Winnie	Kalema	F	winnieK@gmail.com	23-Sep-2000	0
1002	Jacqui	Robert	F	jagi_bob@yahoo.com	9-May-2000	0
1095	Lawrence	Kihara	M	law_k@yahoo.co.uk	25-Dec-2000	0
1909	Veronica	Wafula	F	vero_wafs99@gmail.com	24-Jul-2000	0

Relation Member_Contribution

Field Name	Data Type	Constraints
Cont_No	Number(10)	Primary Key
Cont_Date	Date	Not Null
Cont_Amt	Number(10,2)	Not Null, Check values >= 0
Mem_No	Number(10)	Not Null, Foreign Key from Member

Member_Contribution Data

Cont_No	Cont_Date	Cont_Amt	Mem_ID
101	1-Dec-2001	10,000	907
102	2-Dec-2001	50,000	907
103	2-Dec-2001	9,500	1002
104	3-Dec-2001	15,000	1095

Question ONE

- a) Assume that the above relational structures and data do exist and answer the questions that follow:
- i). What is a database relation? 2 Marks
 - ii). Define the Oracle PL/SQL code to create each of the above relations together with the implied constraints. 6 Marks
 - iii). Define the Oracle PL/SQL code to populate each of the above relations with the provided data. 6 Marks
- b) Define the Oracle PL/SQL code for the query to list the following:
- i). List the names, the email addresses and the account balances of all members who are female and whose last_name ends with the letter "a". 3 Marks
 - ii). List the details of all members that have not made any single contributions. 3 Marks

TOTAL: 20 Marks**Question TWO**

a)

In the historical journey of the development of database technology, many defining and formative defining concepts were embraced. One of these was the American National Standards Institute Standards Planning and Requirements Committee (ANSI-SPARC) Architecture, also known as the Three-Level Architecture.

- i). Being as elaborate and as explicit as possible, describe the Three-Level Architecture. 6 Marks
 - ii). With reference to a commonly used system such as the AMS, describe how the concept of the Three-Level Architecture has informed the creation of different users and the definition of these users' permissions to access data/information that is relevant to them. 5 Marks
- b)
- i). What is normalization? 2 Marks
 - ii). With relevant examples, describe a database transaction. 3 Marks
 - iii). In the development of databases, there were three commonly used data models. Describe each of these data models, being sure to differentiate each from the others. 4 Marks

TOTAL: 20 Marks

Question THREE

Consider the following relation:

Relation X

Mem_No	Mem_Fname	Mem_LName	Cont_No	Cont_Amt
907	Paul	Otiato	101	10,000
907	Paul	Otiato	102	50,000
1002	Jacqui	Robert	103	9,500
1095	Lawrence	Kihara	104	15,000

- a) From the relation above, answer the questions that follow:
- i). With relevant examples, identify the typical relational/update anomalies. 6 Marks
 - ii). Being sure to show all the steps and stating all assumptions you deem necessary, normalize this relation to 3NF. 11 Marks
- b) What are the cardinalities and degrees of the resultant relation in the normalization exercise undertaken above? 3 Marks

TOTAL: 20 Marks

Question FOUR

- a) What is relational algebra? 2 Marks
- b) With the relevant notation, describe each of the following relational algebra operations:
- i). Selection. 3 Marks
 - ii). Projection. 3 Marks
 - iii). Set Difference. 3 Marks
- c) Generate the relational algebra operations expressions together with their Oracle SQL Equivalents:
- i). List details of all members whose first names do not end with letter “e”, and who use a gmail address. 3 Marks
 - ii). List the names of all members. 2 Marks
 - iii). List the names of all members whose first names do not end with letter “e”, and who use a gmail address and who were born after May but before December of any year. 4 Marks

TOTAL: 20 Marks

Question FIVE

- a) With regard to transactions:
- i). Give 2 examples of database transactions. 2 Marks
 - ii). Describe the *acidity* of a transaction. 4 Marks
 - iii). What is a transaction anomaly? Give an example. 3 Marks
- b) Consider the following transaction schedule:

Time	Transaction TRN_200187	Transaction TRN_204932	Balance ACC_Maingi	Balance ACC_Peter
t ₁	Begin Transaction		100,000	10,000
t ₂	Read ACC_Maingi, Read ACC_Peter		100,000	10,000
t ₃	ACC_Maingi = ACC_Maingi - 45,000		100,000	10,000
t ₄	ACC_Peter=ACC_Peter+45,000	Begin Transaction	100,000	10,000
t ₅	Write Bal ACC_Maingi, Write ACC_Peter	Read Bal ACC_Peter	55,000	55,000
t ₆	Abort	Bal ACC_Peter = ACC_Peter - 40,000	100,000	10,000
t ₇	End Transaction	Write Bal ACC_Peter	100,000	15,000
t ₈		Commit	100,000	15,000
t ₉		End Transaction	100,000	15,000

- i). From observing the above transaction schedule, describe what you think is happening. 2 Marks
- ii). With reasons, describe what transaction anomaly has occurred. 3 Marks
- iii). Rewrite the schedule correctly 6 Marks

TOTAL: 20 Marks
